BHUMIK KUMTA

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SUMMARY

Motivated Computer Science undergraduate with a strong foundation in software engineering and growing expertise in AI/ML. Passionate about building scalable, real-world applications using modern technologies.

SKILLS

Languages: Python, C++
Frontend: HTML, CSS, React.js
Backend: Python, Flask
Databases: MySQL, MongoDB
Tools: Git, Docker

EDUCATION

Pimpri Chinchwad University, India
B.Tech – Computer Science Engineering (2023 – 2027)

The Indian School, Bahrain (CBSE)
Class XII – 76%
Class X – 80%

PROJECTS

Feedback Collection System (MERN Stack)
• Built a full-stack feedback management system for collecting and managing user feedback.
• Developed REST APIs and responsive UI for seamless data flow.Tech: MongoDB, Express.js, React.js, Node.js

Iris Classification – End-to-End Machine Learning Classification Project
• Built and compared multiple supervised learning models (Logistic Regression, Random Forest, SVM) to predict Iris flower species with optimized accuracy.
• Conducted exploratory data analysis (EDA), implemented evaluation metrics, and generated visual insights to enhance interpretability of model results.
Tech: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Fake News Detector – End-to-End Machine Learning NLP Project

- Built a production-grade fake news detection system combining a TF-IDF + SGDClassifier pipeline with real-time Google News verification, achieving 98.8% test accuracy and 99.9% ROC-AUC across 5-fold cross-validation.
- Implemented a hybrid verdict engine fusing ML confidence scores with live news similarity (cosine + keyword overlap), making the system robust beyond static training data.
- Integrated LIME explainability to highlight word-level signals driving each prediction, enabling model auditability critical for misinformation detection systems.
- Deployed a FastAPI REST backend with structured logging, Pydantic validation, and a Streamlit frontend; containerized with Docker and managed via docker-compose for one-command reproducibility.

- Established a CI/CD pipeline using GitHub Actions to auto-run pytest test suites and trigger deployment on every push, with predictions logged to a PostgreSQL database for analytics and drift monitoring.
- **Tech: Python, Scikit-learn, FastAPI, Streamlit, Docker, PostgreSQL, GitHub Actions, LIME, Pandas, NumPy**